Operations Research

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Exercise Sheet 4

Training Exercises

Exercise T4. 1. *Duality & Complementary Slackness*

a) Derive the dual LP from the following LP.

$$\begin{array}{lrcl}
(P)\ \underset{\text{s.t.}}{\text{Min}} & 2x_1 + 2x_2 + 2x_3 - x_4 & & \\
& -x_1 + x_2 + 2x_3 - x_4 & \geq & 15 \\
& x_1 - x_2 + 4x_4 & = & 5 \\
& 2x_2 - 2x_3 + x_4 & \leq & -15 \\
& x_1, x_2, x_3 & \geq & 0 \\
& x_4 & \leq & 0
\end{array}$$

b) Let $x^T = (5, 0, 10, 0)$ be a primal point. Use the complementary slackness conditions to check whether x is an optimal solution for (P).

Exercise T4. 2. *Duality*

Consider the following linear programs, with $A \in \mathbb{R}^{n \times n}$ and $b, c \in \mathbb{R}^n$.

$$\begin{array}{llll}
(P_1)\ \underset{\text{s.t.}}{\max} & c^T x & & \\
& Ax & = b & \\
& x & \in \mathbb{R}^n_{\geq 0} &
\end{array}
\qquad
\begin{array}{llll}
(P_2)\ \underset{\text{s.t.}}{\min} & c^T x & & \\
& Ax & = b & \\
& x & \in \mathbb{R}^n_{\geq 0} &
\end{array}$$

$$\begin{array}{llll}
(P_3)\ \underset{\text{s.t.}}{\min} & b^T x & & \\
& A^T x & \geq c & \\
& x & \in \mathbb{R}^n &
\end{array}
\qquad
\begin{array}{llll}
(P_4)\ \underset{\text{s.t.}}{\max} & b^T x & & \\
& A^T x & = c & \\
& x & \in \mathbb{R}^n &
\end{array}$$

Let z_1, z_2, z_3 and z_4 be the optimal objective function values of $(P_1), (P_2), (P_3)$ and (P_4) respectively.

a) Determine the corresponding dual linear program for each of the above programs. Identify primal-dual pairs among the linear programs $(P_1) - (P_4)$.

b) Show that if $(P_1), (P_2), (P_3), (P_4)$ are feasible, then $z_1 = z_2 = z_3 = z_4$.

c) If $z_1 > z_2$, which linear program cannot be feasible? Explain your answer.

Exercise T4. 3. *Rock-Paper-Scissors*

George and Max play rock-paper-scissors. The rules are simple: both players simultaneously choose a symbol from the set {Scissors, Rock, Paper}. Scissors beats paper, paper beats rock and rock beats scissors. If both players choose the same sybmol, the round ends in a draw. The winner of each round receives 2 euros from the loser. In the case of a draw, no money is exchanged.

a) Build the payoff matrix of this game for Max and George.

b) Model the problem of finding an optimal strategy for George and Max as a linear program. State the optimal strategy and the win/loss for both players.

c) Max proposes the following change to George. In the event of a tie, George receives 1.5 euros from Max if rock or scissors was played. If paper was played, Max receives 3 euros from George. Determine the optimal strategies for this modified problem. Should George accept the proposal? Justify your answer.

Hint: Use Gurobi to solve the LPs.

Self-study Exercises

Exercise S4. 1. *Duality*

$$
\begin{array}{lrcrcr}
(P) & \underset{\text{s.t.}}{\text{Max}}\ x_1 & + & x_2 & & \\
 & -3x_1 & + & 2x_2 & \le & -1 \\
 & x_1 & - & x_2 & \le & 2 \\
 & x_1, & & x_2 & \ge & 0
\end{array}
$$

a) Formulate the corresponding dual LP to (P)

b) List all complementary slackness conditions of these LPs.

c) Does the primal LP have feasible solutions? Justify your answer.

d) Is the primal LP bounded? Justify your answer.

Exercise S4. 2. *Duality & Complementary Slackness*

a) Derive the dual LP (D) from the following LP:

$$
\begin{array}{llrcr}
(P) & \underset{\text{s.t.}}{\text{Max}} & 15x_1 + 5x_2 - 15x_3 & & \\
 & & -x_1 + x_2 & \le & 2 \\
 & & x_1 - x_2 + 2x_3 & \le & 2 \\
 & & 2x_1 - 2x_3 & \le & 2 \\
 & & -x_1 + 4x_2 + x_3 & \ge & -1 \\
 & & x_1 & \ge & 0 \\
 & & x_3 & \le & 0
\end{array}
$$

b) State all complementary slackness conditions.

c) Let $y^\tau = (9, 0, 16, -1)$ be a dual point. Use the complementary slackness conditions to check whether y is an optimal solution for (D).

d) Let $x^\tau = (1, 3, 0)$ be a primal point. Use the complementary slackness conditions to check whether x is an optimal solution for (P).